Mahavir Education Trust's

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SAKEC OBE MANUAL VERSION 3.0 – ASSESSMENT METHODOLOGY & RUBRICS DOSSIER

Course: Cloud-Native Distributed Systems & Microservices (22CS501) • Academic Year: 2025-26 • Semester: V

1. Comprehensive Assessment Architecture

Component	Code	Allocated Marks	Pedagogical Methodologies Selected
Continuous Internal Assessment (Theory)	**CIA**	**20 Marks**	Technical Reel Explainer (60s), Cloud Outage Case Study Analysis
Mid-Semester Examination (Theory)	**MSE**	**30 Marks**	Higher-Order Thinking Questions (HOTS), System Bottleneck Scenario Paper
Continuous Internal Assessment for Practical / Lab	**CIAP**	**25 Marks**	Continuous Lab Experiment Audit, Live Code Debugging Challenge
End-Semester Examination (Theory)	**ESE**	**50 Marks**	Comprehensive Theory Exam Blueprint, Bloom's Balanced Part A & B Question Paper
End-Semester Practical & Oral Examination	**ESEP**	**25 Marks**	Hands-on System Demonstration, External Viva-Voce Examination
Self-Learning Assessment (SLA)	**SLA**	**20 Marks**	NPTEL / MOOC Certification Audit, Capstone Open-Source Mini-Project
Total Assessment	**TOTAL**	**170 Marks**	**Aggregate Passing Criterion: 40% Marks**

2. Component-Wise Activity Guidelines & Student Deliverables

CIA: Continuous Internal Assessment (Theory) **Marks: 20** **Target COs: CO1, CO2, CO3**

Bloom's Level Focus:	Cognitive: L3 (Apply) & L4 (Analyze)
Activity Description:	Selected OBE Activities: Technical Reel Explainer (60s), Cloud Outage Case Study Analysis. Faculty Scenario Instructions: Require students to record a 60-second video explaining consensus for CIA 1, and conduct an AWS outage case study for CIA 2.
Faculty Guidelines:	Administer according to SAKEC OBE Manual Section 25.3-25.8. Share Microsoft Teams rubric with students prior to submission. Evaluate using the 4-level analytic scale (Excellent 4, Good 3, Satisfactory 2, Needs Improvement 1).
Student Deliverables:	Activity artifact (60-90s technical reel / case study report / presentation deck) uploaded directly to Microsoft Teams Assignment.

3. AICTE 4-Level Performance Evaluation Rubrics

Rubric for CIA (Continuous Internal Assessment (Theory)) Rubric for MSE (Mid-Semester Ex-

MSE: Mid-Semester Examination (Theory)	Marks: 30	Target COs: CO1, CO2, CO3
Bloom's Level Focus:	Cognitive: L2 (Understand) to L4 (Analyze)	
Activity Description:	Centralized written examination evaluating foundational modules covered in the first half of the semester. Every question is mapped explicitly to target Course Outcomes (COs) and Bloom's cognitive levels (RBT).	
Faculty Guidelines:	Prepare question-wise CO mapping matrix and model answer key. Record question-wise student marks to compute percentage scoring ¿= 60%.	
Student Deliverables:	Written examination script solving analytical and scenario-based engineering problems.	

CIAP: Continuous Internal Assessment for Practical / Lab	Marks: 25	Target COs: CO2, CO4, CO5
Bloom's Level Focus:	Psychomotor (P2-P3), Cognitive (L3-L4) & Affective (A3)	
Activity Description:	Weekly laboratory experiment evaluation and automated test-case verification (Continuous Lab Experiment Audit, Live Code Debugging Challenge). Assesses Conceptual Understanding (20%), Execution (15%), Debugging (20%), Professional Conduct & Ethics (20%), Output Accuracy (15%), and Documentation (10%).	
Faculty Guidelines:	Evaluate weekly laboratory experiments using SAKEC Annexure B Section 25.1 rubrics (Computer-Allied / Circuit-Allied / PARIKSHAK).	
Student Deliverables:	Executed code/circuit, passed test cases, validated output, and structured laboratory report.	

ESE: End-Semester Examination (Theory)	Marks: 50	Target COs: CO1, CO2, CO3, CO4, CO5, CO6
Bloom's Level Focus:	Comprehensive Cognitive: L2 to L6	
Activity Description:	Comprehensive end-semester theory examination covering all curriculum modules (SAKEC 60% University Assessment component of Direct Attainment).	
Faculty Guidelines:	Ensure balanced coverage of all Course Outcomes and Bloom's Taxonomy levels across the question paper.	
Student Deliverables:	Comprehensive written answer script.	

ESEP: End-Semester Practical & Oral Examination	Marks: 25	Target COs: CO4, CO5, CO6
Bloom's Level Focus:	Psychomotor (P4 Articulation) & Cognitive (L5-L6)	
Activity Description:	Practical and oral viva-voce examination conducted at the end of the semester to evaluate practical skills and application of knowledge.	
Faculty Guidelines:	Joint evaluation by Internal Examiner and External Examiner using structured laboratory performance and viva criteria.	
Student Deliverables:	Live practical execution, system demonstration, and oral defense.	

SLA: Self-Learning Assessment (SLA)	Marks: 20	Target COs: CO5, CO6
Bloom's Level Focus:	Cognitive (L5-L6) & Affective (A3-A4)	
Activity Description:	Semester-long Self-Learning Activity evaluated across 3 progressive phases: Phase 1: Problem Understanding & Design Methodology (25%), Phase 2: Methodology Validation & Implementation (25%), Phase 3: Comprehensive Evaluation & Trade-offs (20%), Final Report Quality (15%), Plagiarism & AI Compliance (10%), Defense (5%).	
Faculty Guidelines:	Conduct progressive reviews at Phase 1, Phase 2, and Phase 3 per SAKEC OBE Manual Section 25.2. Verify AI usage disclosure and plagiarism compliance.	
Student Deliverables:	Phase I, II, III progressive review artifacts, final comprehensive technical report, and oral defense.	

amination (Theory)) Rubric for CIAP (Continuous Internal Assessment for Practical / Lab) Rubric for ESE (End-Semester Examination (Theory)) Rubric for ESEP (End-Semester Practical & Oral Examination) Rubric for SLA (Self-Learning Assessment (SLA))

Evaluation Criterion	Exemplary (Level 4)	Proficient (Level 3)	Developing (Level 2)	Novice (Level 1)
Clarity of Concept (30%)	Concept is clearly explained with accurate examples and appropriate interpretation.	Concept is clearly explained with relevant information.	Concept explanation is inadequate or lacks clarity.	Concept explanation is incorrect or demonstrates misunderstanding.
Creativity & Engagement (20%)	Highly creative, visually appealing, and strongly engaging presentation.	Creative approach with moderate engagement and visualization.	Basic visualization with limited creativity.	Dull or ineffective visualization with poor engagement.
Technical Content (Simulation/Logic) (20%)	Relevant code and logic are correctly used with proper comments, structure, and indentation.	Correct code and logic are used but comments or indentation require improvement.	Code addresses some conditions but does not completely satisfy all requirements.	Poor code quality; code does not execute or fails to address basic conditions.
Communication & Presentation (15%)	Smooth, confident, and audience-friendly communication.	Clear communication with minor hesitation.	Basic effort in communication and presentation.	Low effort with ineffective communication.
Real-world Connection / Metaphor (15%)	Strong metaphor or real-world scenario is used to explain the concept effectively.	Good connection with a scenario, but some conditions or scope limitations are missing.	Basic connection is demonstrated but scope is unclear.	Connection is irrelevant or missing.

Evaluation Criterion	Exemplary (Level 4)	Proficient (Level 3)	Developing (Level 2)	Novice (Level 1)
Understanding of Scenario/Problem Context (20%)	Demonstrates clear understanding of the given scenario/problem; correctly identifies key issues, constraints, and objectives.	Demonstrates good understanding; identifies most relevant aspects of the scenario.	Shows basic understanding; identifies some key elements.	Shows minimal understanding; fails to identify key elements of the scenario.
Analysis & Application of Concepts (30%)	Applies relevant technical/engineering knowledge to analyze the scenario; demonstrates logical reasoning and may include diagrams, calculations, or references.	Applies mostly relevant technical knowledge with minor errors or gaps.	Applies basic or partially relevant concepts; analysis is present but limited in clarity.	Analysis is weak, incomplete, or mostly incorrect.
Solution / Decision-Making & Justification (30%)	Provides a feasible and well-thought-out solution; justifies the approach using data, theory, or best practices.	Provides a valid solution; justification lacks strong supporting evidence.	Provides a basic or partially feasible solution; justification is weak.	Provides a vague, unclear, or impractical solution with little to no justification.
Report / Assignment Quality (20%)	Clear, concise, and well-organized report demonstrating effective presentation of ideas.	Report is generally well-written but may lack clarity or organization.	Report lacks clarity or organization.	Poorly written report with inadequate presentation.

4. Course Outcome Attainment Calculation Formulation

Evaluation Criterion	Exemplary (Level 4)	Proficient (Level 3)	Developing (Level 2)	Novice (Level 1)
Conceptual Understanding (Cognitive) (20%)	Clearly explains objectives, theory, algorithms, tools and expected outcomes with strong clarity.	Explains objectives and concepts correctly with minor gaps.	Partial understanding; relies on lab manual or instructions.	Unable to explain purpose or underlying concept.
Execution (Psychomotor) (15%)	Independently uses IDE, libraries and tools; code is correct, efficient and well-structured.	Executes code correctly with minor errors; minimal guidance required.	Frequent syntax/logical errors; needs repeated assistance.	Unable to execute experiment even with help.
Problem-Solving & Debugging (Cognitive + Psychomotor) (20%)	Systematically identifies errors, debugs efficiently and optimizes solution.	Identifies errors and fixes them with guidance.	Struggles to locate or resolve errors.	No meaningful attempt to debug.
Professional Conduct & Lab Ethics (Affective) (20%)	Highly disciplined, punctual, ethical; follows academic integrity and lab rules.	Generally disciplined with minor lapses.	Inconsistent discipline or over-dependence on peers.	Unprofessional behaviour; violates lab norms.
Output / Result Accuracy (Product) (15%)	Output fully correct, validated and meets experiment objectives.	Output mostly correct with minor deviations.	Partially correct output.	Incorrect or no output.
Documentation & Result Analysis (Cognitive) (10%)	Well-structured report with algorithm, code, screenshots, results and insightful analysis.	Complete documentation with basic analysis.	Incomplete or poorly explained report.	Very weak or missing documentation.

Evaluation Criterion	Exemplary (Level 4)	Proficient (Level 3)	Developing (Level 2)	Novice (Level 1)
Understanding of Scenario/Problem Context (20%)	Demonstrates clear understanding of the given scenario/problem; correctly identifies key issues, constraints, and objectives.	Demonstrates good understanding; identifies most relevant aspects of the scenario.	Shows basic understanding; identifies some key elements.	Shows minimal understanding; fails to identify key elements of the scenario.
Analysis & Application of Concepts (30%)	Applies relevant technical/engineering knowledge to analyze the scenario; demonstrates logical reasoning and may include diagrams, calculations, or references.	Applies mostly relevant technical knowledge with minor errors or gaps.	Applies basic or partially relevant concepts; analysis is present but limited in clarity.	Analysis is weak, incomplete, or mostly incorrect.
Solution / Decision-Making & Justification (30%)	Provides a feasible and well-thought-out solution; justifies the approach using data, theory, or best practices.	Provides a valid solution; justification lacks strong supporting evidence.	Provides a basic or partially feasible solution; justification is weak.	Provides a vague, unclear, or impractical solution with little to no justification.
Report / Assignment Quality (20%)	Clear, concise, and well-organized report demonstrating effective presentation of ideas.	Report is generally well-written but may lack clarity or organization.	Report lacks clarity or organization.	Poorly written report with inadequate presentation.

Evaluation Criterion	Exemplary (Level 4)	Proficient (Level 3)	Developing (Level 2)	Novice (Level 1)
Test Case Pass Percentage (Cognitive + Psychomotor, Auto) (60%)	Passes all test cases including edge and hidden cases.	Passes 75.00%-99.99% test cases.	Passes 50%-74.99% test cases.	Passes less than 50% test cases or runtime error.
Code Structure & Programming Practices (Psychomotor, Manual) (20%)	Modular, readable code; proper indentation, naming and comments.	Mostly readable code with minor style issues.	Poor structure; limited readability.	Disorganized, unreadable or copied code.
Professional Conduct & Academic Integrity (Affective, Manual) (20%)	Original work; ethical conduct; no plagiarism.	Minor guidance used; integrity maintained.	Over-dependence on peers/resources.	Plagiarized or unethical submission.

Evaluation Criterion	Exemplary (Level 4)	Proficient (Level 3)	Developing (Level 2)	Novice (Level 1)
Problem Understanding & Design Methodology Formulation (Phase 1) (25%)	Clear problem framing; well-structured, justified methodology.	Correct understanding with minor gaps.	Partial understanding; generic methodology.	Weak or incorrect methodology.
Methodology Validation - Implementation / Analytical Evidence (Phase 2) (25%)	Strong validation using implementation or analytical evidence.	Adequate validation with minor issues.	Limited or superficial validation.	Validation missing or incorrect.
Methodology Evaluation, Trade-offs & Decision Rationale (Phase 3) (20%)	Rigorous evaluation; justified decisions across scenarios and constraints.	Reasonable evaluation with limited scenarios.	Narrow or weak evaluation.	Arbitrary or unsupported decisions.
Final Comprehensive Report Quality (15%)	Well-organized, coherent, technically sound report.	Minor clarity or structure issues.	Basic but acceptable report.	Incomplete or poorly structured.
Plagiarism & AI Usage Compliance (10%)	Low similarity; transparent and ethical AI declaration.	Acceptable similarity; partial AI disclosure.	High similarity or unclear disclosure.	Plagiarism or unethical AI use.
Review Interaction & Defense (5%)	Confident, evidence-based justification.	Minor gaps in defense.	Limited clarity.	Unable to justify.

NBA Direct Attainment Formula

Per SAKEC OBE Manual Version 3.0 (Section 17 & 20): Expected Proficiency threshold is fixed at 60% marks. Let X = Percentage of students scoring ¿= 60% marks. If X ¿= 70%, Attainment Level = 3.00. If 60% ¡= X ¡ 70%, linear interpolation applies: Attainment Level (y) = 0.2 * X - 11. If X ¡ 60%, Attainment Level = X / 60. Final Course Outcome Attainment = (0.80 * Direct CO Attainment) + (0.20 * Indirect CO Attainment via Exit Survey). Initial CO Attainment Target for new courses is 1.80.

Course Coordinator (Faculty)	**Head of Department (HOD)**	**Dr. Bhavesh Patel (Principal)**
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